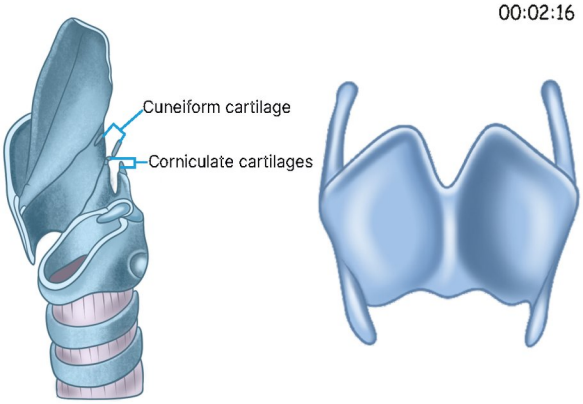
17. LARYNX

CARTILAGES OF LARYNX

00:02:16



- Larynx is made of cartilages
- Largest- Thyroid cartilage
  - Has 2 ala like a butterfly
    - → Meet at an angle of 90°- Adams apples
    - → 120° in females
  - Muscles attached to the oblique line
    - → Sternothyroid
    - → Thyrohyoid
    - → Thyropharyngeus
- Below the Thyroid cartilage- Cricoid cartilage (Signet ring cartilage)
- Inner surface of thyroid epiglottic cartilage
- Over the superior facet of the cricoid cartilage- Arytenoid
- Above the arytenoid cartilage- Cuneiform cartilage

Paired cartilages	Unpaired cartilages
• Arytenoid • Corniculate • Cuneiform	• Thyroid • Cricoid • Epiglottis

- Some cartilages calcify with age, called hyaline cartilage
  - Thyroid
  - Cricoid
  - Arytenoid
- Elastic cartilage opens and closes the laryngeal inlet
  - Epiglottis
  - Circulate
  - Cuneiform
- The only complete cartilaginous ring in the airway is the cricoid cartilage

Cricoid Cartilage

00:06:22

- Only signet ring-shaped cartilage
- The narrow arch anteriorly, & the lamina is wider
- On line 2 facets
  - Superior facets- Attachment to the arytenoid cartilage
  - Inferior facets- Attachment to the inferior horn of the thyroid cartilage
- Laryngeal crepitus can be seen
  - Movement of the cricoid cartilage against the vertebra- Gritty sensation felt
  - Normally present in everyone

YOURWISH

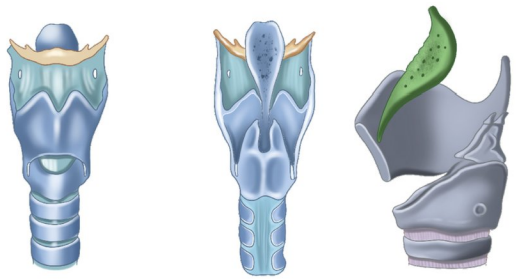
- Absent laryngeal crepitus- Boccas's sign seen in
  - → Post-cricoid cancer
  - → Retropharyngeal abscess
  - → Prevertebral abscess

Epiglottic Cartilage

00:08:56

FMGE 2021

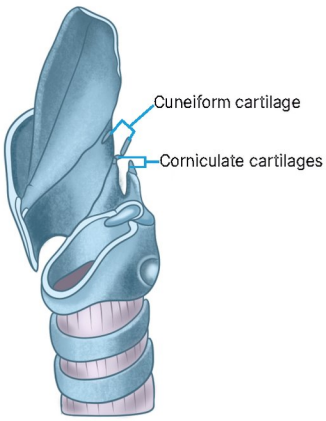
- Leaf-like cartilage is attached to the inner surface of the Thyroid cartilage
- Surfaces
  - Suprahyoid
  - Infrahyoid
- Surface that faces the base of the tongue- Lingual surface
- Surface that faces the larynx- Laryngeal surface
- Superiorly free
- Attached inferiorly to the thyroid cartilage

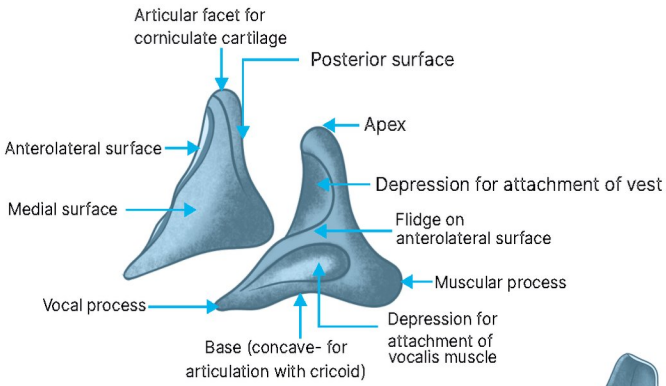


Arytenoid Cartilage

INICET 2020

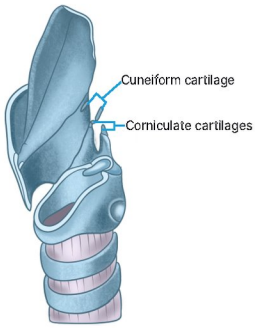
- Pyramid-shaped cartilage
- Base rests on the superior facet of the cricoid cartilage
- Apex- Cornciculate, cuneiform cartilage
- Medially, a vocal process is present that attaches to the vocal cords
- Muscular process that provides attachment to the muscle



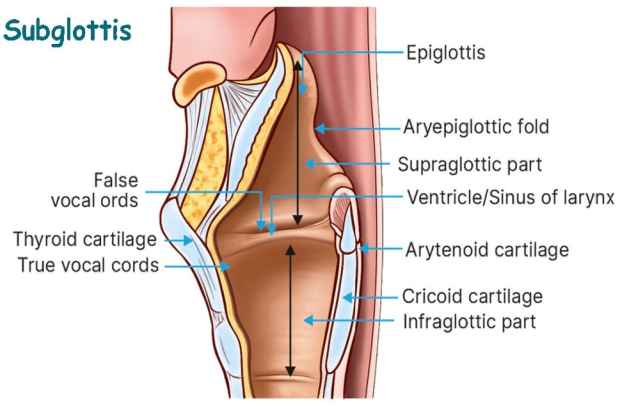


Corniculate And Cuneiform Cartilage

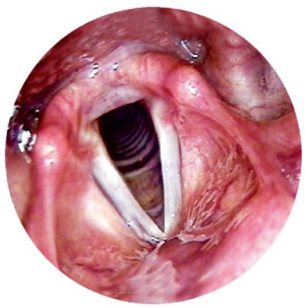
- Corniculate, AKA Cartilage of Santorini
- Cuneiform cartilage, AKA Cartilage of Wrisberg



Glottis, Supraglottis & Subglottis



Sagittal section of larynx



YOURWISH

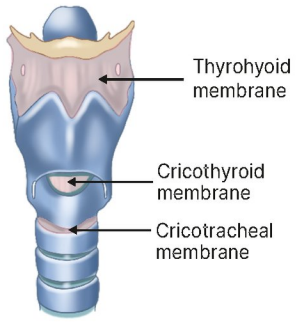
- The larynx is classified into 3 subdivisions
- Glottis- At the level of the vocal cords and 1 cm below
- Supraglottis- Above the level of the vocal cords
- Subglottis - 1 cm below the vocal cord to the lower border of the Cricoid cartilage

Development

00:12:44

- Supraglottis develops from the 4th arch
- Glottis & Subglottis develop from the 6th arch

MEMBRANES OF LARYNX

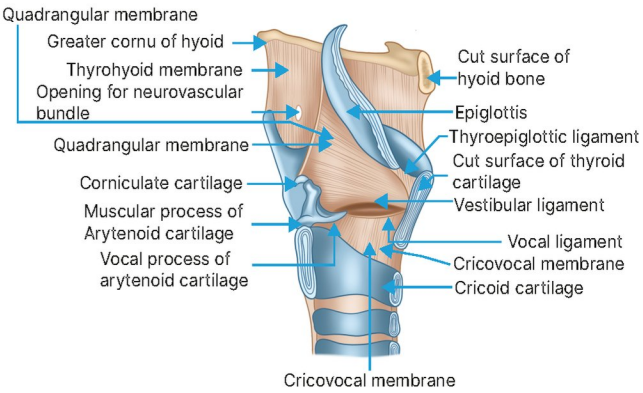


Extrinsic

- Connects the laryngeal structure to an external structure
  - Thyrohyoid membrane- SLN nerve block
  - Cricotracheal membrane

Intrinsic

- Connects the laryngeal structure to another laryngeal structure
- Cricothyroid membrane- Emergency airway access
- Quadrangular membrane
  - Has 4 sides
  - Extends between the epiglottis, thyroid, and arytenoid cartilages
  - Lower fibres condense to form false cords (Vestibular ligament originates)
- Cricovocal
  - Cone-shaped, AKA Conus elasticus
  - Gives rise to the vocal ligament
  - Upper fibres condense to form true cords



LARYNGEAL CAVITY

00:18:37

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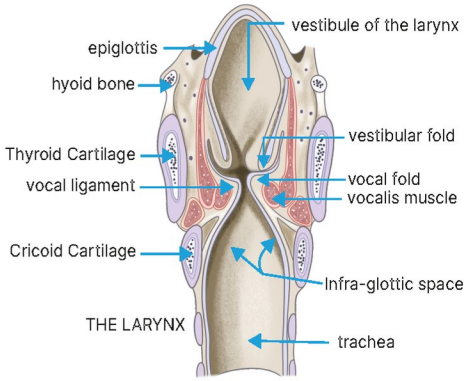
- Superiorly begins- Inlet of larynx
- Quadrangular membrane
- Below cricovocal membrane
- Space between 2 quadrangular membranes- Vestibule of the larynx
- Space between 2 false chords- Rima vestibuli
- Space between 2 true chords- Rima glottis



YOURWISH

- Space between the false and true chords- Ventricle of the larynx
- Below the rima glottis- Subglottis

EPITHELIUM OF THE LARYNX

00:21:45

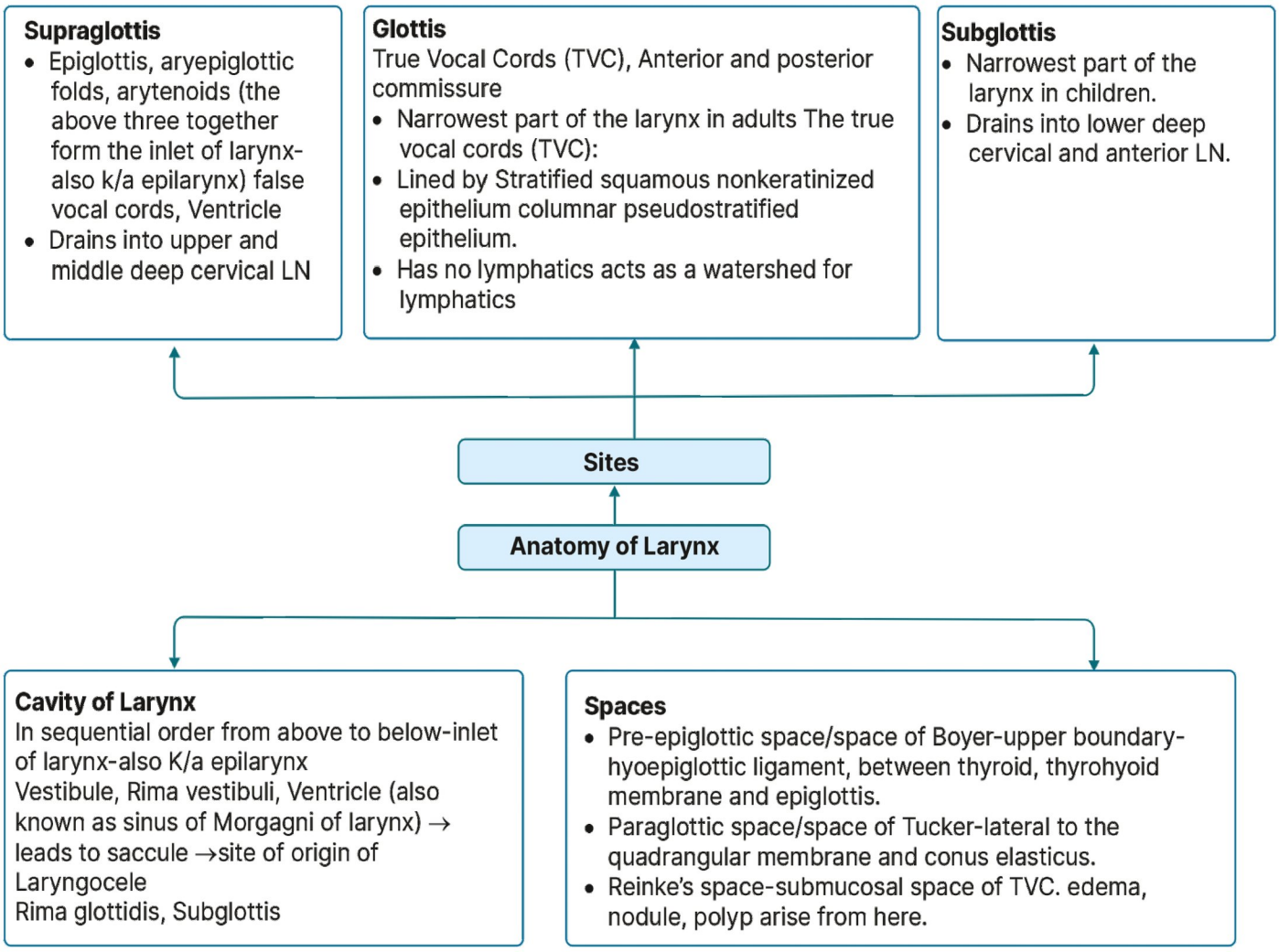
- Lined by pseudostratified ciliated columnar epithelium
- Except for 3 sites
  - Vocal cords
  - Aryepiglottic fold
  - Superior surface of the epiglottis

LYMPHATICS

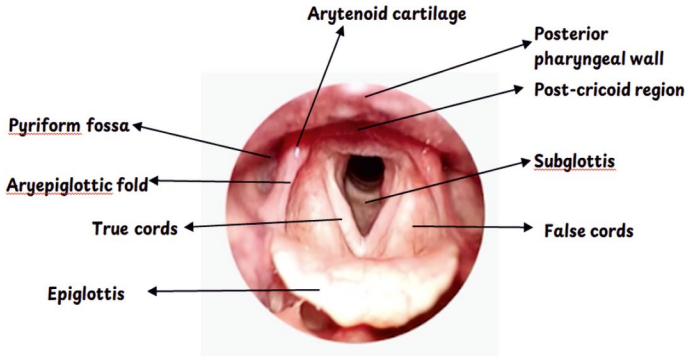
00:22:35

- Supraglottis- Drains into the upper deep cervical lymph nodes
- Subglottis- Pre-laryngeal and pre-tracheal lymph nodes into lower deep cervical lymph nodes or mediastinal nodes
- Glottis- No lymphatics

Summary



Endoscopic View Of The Larynx

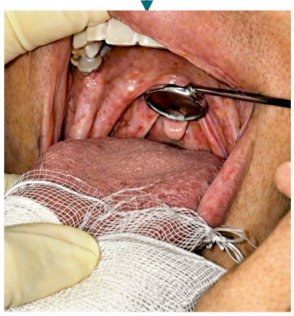


METHODS OF EXAMINING THE LARYNX

00:27:24

- Indirect laryngoscopy- Mirror placed behind the uvula
- Direct laryngoscopy
- Rigid endoscopy with 70°
- Fibreoptic laryngopharyngeoscopy
- Stroboscopy- Mucosal wave can be seen
  - Absent mucosal wave- Carcinoma & Cyst

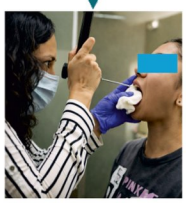
Indirect laryngoscopy



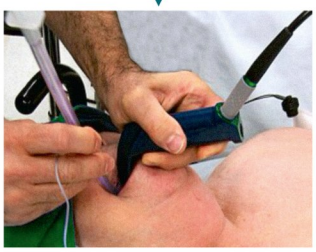
Endoscopy with or without stroboscope (visualization of mucosal wave in slow motion to differentiate lesions and site of involvement)

Flexible endoscopy

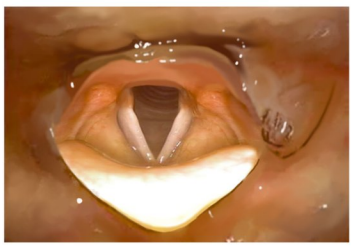
Rigid Endoscopy
Tips of different angles (70-90)

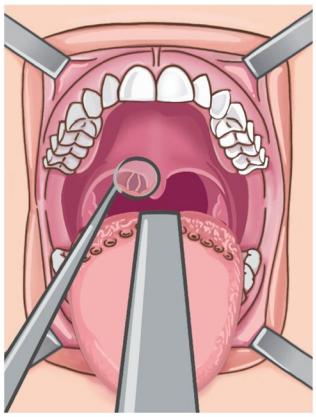


Direct laryngoscopy



Boyce position
Chevalier Jackson/Barking dog/sniffing morning air-
Flexion at cervical spine and extension at atlanto-occipital joint.





MUSCLES OF THE LARYNX

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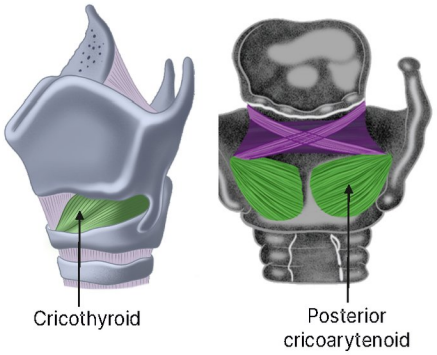
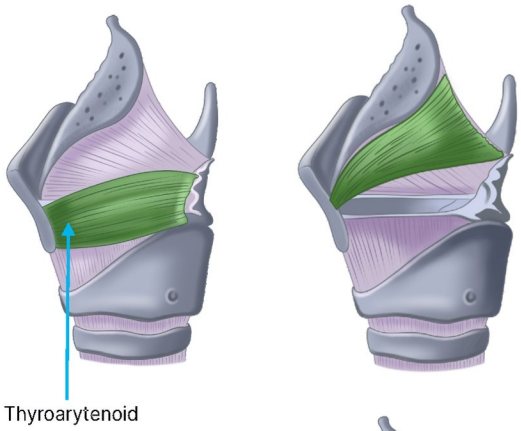
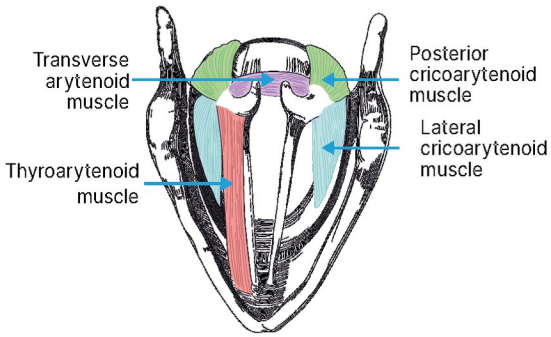
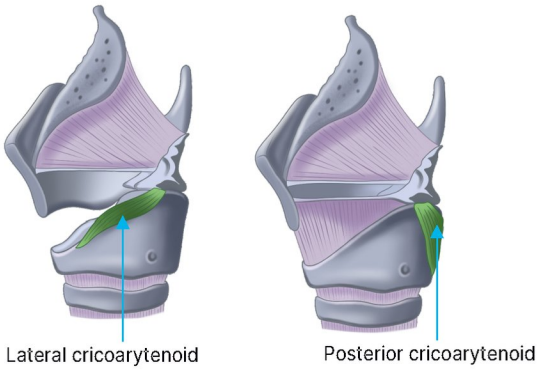
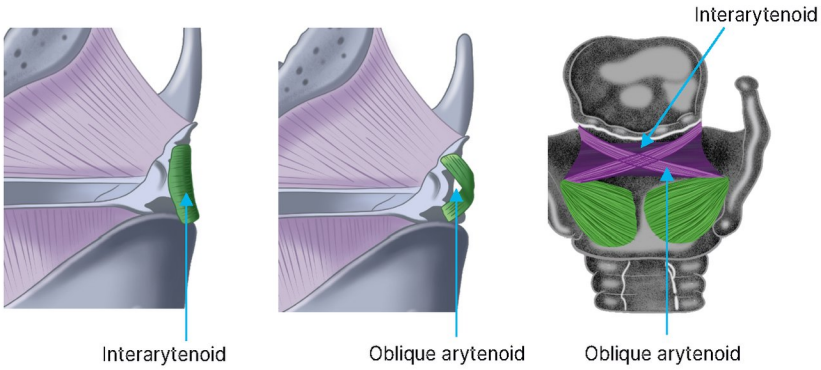
- Abductor of the vocal cord/safety muscle of the larynx- Posterior cricoarytenoid
- Tensor of the vocal cord/secondary adductor- Cricothyroid
  - Has 2 bellies
  - Supplied by the external branch of SLN
- Adduct- All muscles of the larynx except the posterior cricoarytenoid
- Adductor + tensor- Cricothyroid and vocalis
- Increase tension/pitch/length- Cricothyroid (only intrinsic muscle lying outside)
- Decrease tension /pitch/length- Vocalis
- Only unpaired- Interarytenoid/transverse arytenoid
  - Receives dual innervation











SPACES OF LARYNX

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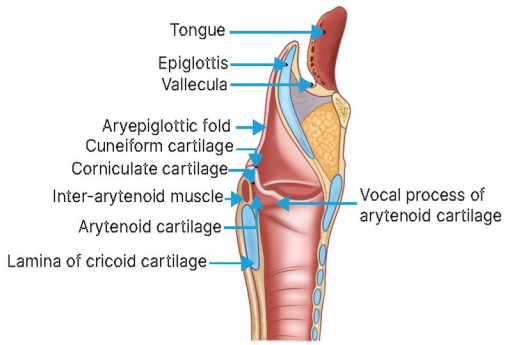
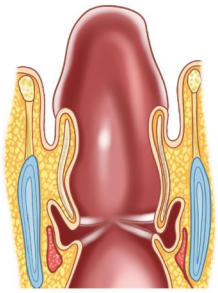
- Preepiglottic space of Boyer
- Paraglottic space of Tucker
- Reinke's space

YOURWISH

YOURWISH

PREIPIGLOTTIC SPACE OF BOYER
 
 	
	• Space is in front of the epiglottis
	• Boundaries
	  ○ Anteriorly- Thyrohyoid membrane
	  ○ Superiorly- Hyoepiglottic ligament
	  ○ Posteriorly- Epiglottis

PARAGLOTTIC SPACE OF TUCKER
 
	• Lies on the sides of the larynx
	• Boundaries
	  ○ Laterally by the Thyroid cartilage
	  ○ Medially Quadrangular membrane
	  ○ Below- Cricovocal membrane and ventricles

REINKE'S SPACE	• There are 4 layers of the vocal cords
	  ○ Epithelium
	  ○ Superficial lamina propria
	  ○ Vocal ligament
	  ○ Vocalis muscle
	• The space between the epithelium and the superficial lamina propria is called Reinke's space

FUNCTIONS OF THE LARYNX

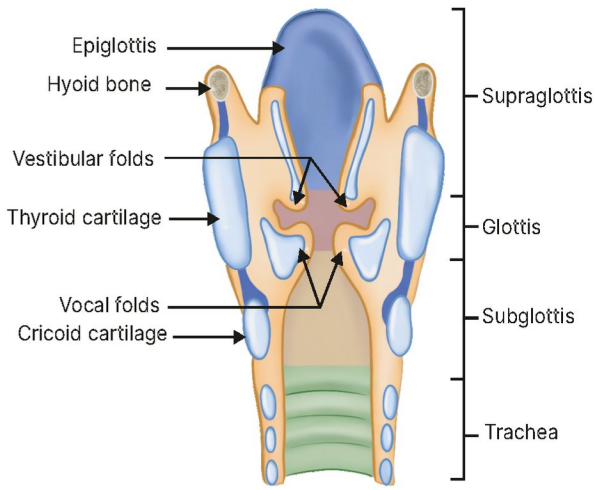
00:44:44

- Protection of airways
- Phonation
- Respiration
- Fixation of the chest

	INFANTS	ADULTS
POSITION	C2-C3	C3-C6
EPIGLOTTIS	Curled	Leaf shaped
THYROID RTILAGE	Flat	Angulated
NARROWEST PART	Subglottis	Glottis
SUBMUCOSAL TISSUE	More	Less
CARTILAGES	Soft	Ossified

ONE LINERS

- Larynx develops from the 4th and 6th arches
- Supraglottis develops from the 4th arch
- Glottis and subglottis develop from the 6th arch
- Larynx lies against C3-C6
- Laryngeal crepitus occurs due to the movement of the cricoid against the vertebrae
- Laryngeal crepitus is absent in post-cricoid cancer, prevertebral abscess, and retropharyngeal abscess
- Hyaline cartilages in the larynx are the thyroid, cricoid, and arytenoid
- Elastic cartilages is epiglottis, cuneiform, and corniculate
- Largest cartilage in the larynx thyroid
- Signet ring cartilage is cricoid
- Angle at which the thyroid meets the ala
  - In males- 90°
  - In female- 120°
- Oblique line of the thyroid cartilage gives attachment to the sternothyroid, thyrohyoid, and thyropharyngeus
- 2 surfaces of the epiglottis- lingual and laryngeal surfaces
- Parts of the epiglottis
  - Suprhyoid
  - Infrahyoid
- Only complete cartilaginous ring in the entire airway cricoid cartilage
- Narrowest portion of the adult larynx is the glottis, and in the paediatric larynx is the subglottis
- Cartilage of Santorini is Corniculate
- Cartilage of Wrisberg is Cuneiform
- Membranous portion is 2/3rd, and the cartilaginous portion is 1/3rd
- Sinus of Morgagni of the larynx-ventricles
- Rima vestibuli is the space between the false cord
- Rima glottidis is the space between the true cord
- Space of Tucker is a paraglottic space
- Space of Boyer is the pre-epiglottic space
- Only unpaired laryngeal muscle is the interarytenoid/transverse arytenoid
- Only muscle that receives dual innervation is the interarytenoid/transverse arytenoid
- Only abductor of the vocal cord is the postcricoid arytenoid
- Tensor is cricothyroid



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COMMON ERRORS

- Nerve supply
  - Sensory above the vocal cords- Internal branch SLN
  - Below the vocal cord- RLN
  - Motor supply- All muscles by RLN except cricothyroid, which is supplied by the external branch of SLN
- Muscles
  - Cricothyroid- Only muscle with 2 bellies lying outside- Adducts and lengthens the vocal cord
  - Vocalis- shortens and adducts the vocal cord
  - Posterior cricoarytenoid- Only abductor of the vocal cord

CONGENITAL ANOMALIES OF THE LARYNX

- M/c Congenital Anomaly of Larynx:
  - 1st m/c = Laryngomalacia
  - 2nd m/c = Congenital Vocal Cord Paralysis
  - 3rd m/c = Subglottic Stenosis

LARYNGOMALACIA

00:51:44

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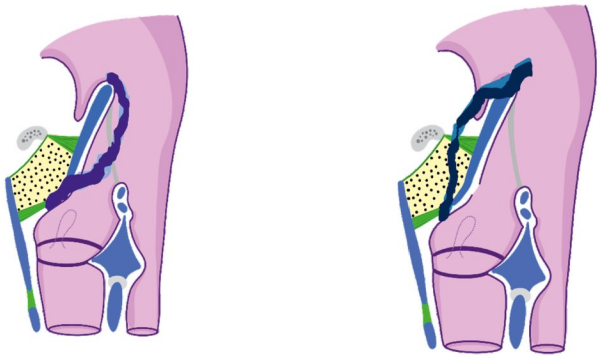
- Definition: A condition where there is excessive flaccidity of the epiglottis or supraglottis
  - Supraglottis: Part of the larynx that lies above the vocal cords

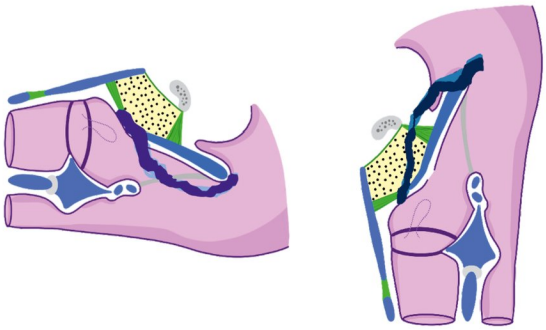
Mechanism Of Obstruction



- During inspiration:
  - Air gets sucked in
  - Epiglottis falls on the airway → obstructs the airway
  - Results in inspiratory stridor
- During expiration:
  - Air comes from below (from the trachea)
  - Pushes the epiglottis away
  - No stridor during expiration

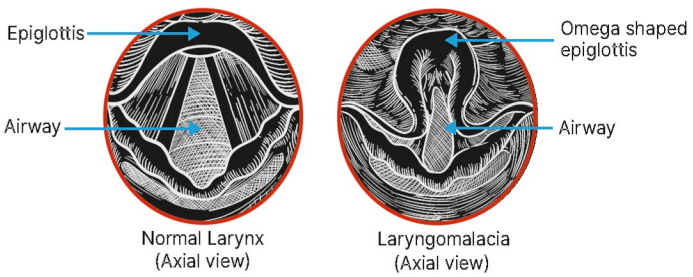
Clinical Findings & Presentation

- Characteristics of Stridor:
  - Inspiratory stridor at birth
  - No stridor during expiration
- Effect of crying on stridor: Increases stridor
  - Crying → negative intrathoracic pressure → Pulling air inside → epiglottis gets sucked in → increased stridor
  - Cry/voice at birth: Normal
    → Because epiglottis are involved, not the vocal cords
- Effect of Position
  - Supine position:
    → Epiglottis collapses and falls over the laryngeal inlet → obstruction to the glottis → worsens stridor
  - Prone position:
    → Epiglottis falls away from the larynx → relief of stridor



Laryngoscopic Examination

- Examination Method:
  - Done under short general anesthesia or sedation
  - Laryngomalacia Finding:
    → Epiglottis have collapsed
    → Omega-Shaped Epiglottis



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Treatment

- Conservative: Wait and watch
  - Spontaneous resolution happens at 1-2 years of age
  - May need epiglottic surgeries if not resolved

CONGENITAL VOCAL CORD PALSY

- M/c congenital CNS abnormality causing vocal cord palsy: Arnold-Chiari Malformation

Clinical Presentation

- Unilateral Vocal Cord Paralysis
  - Usually asymptomatic, or may have a weak cry
- Bilateral Vocal Cord Paralysis
  - Stridor at birth
    → Biphasic stridor (because of a lesion at the level of the vocal cords)
  - Difficulty in breathing
  - Cyanosis
  - Respiratory distress at birth
  - Cry/Voice:
    → Weak cry
    → Aphonia or no cry at all
  - Effect of Crying: increases stridor

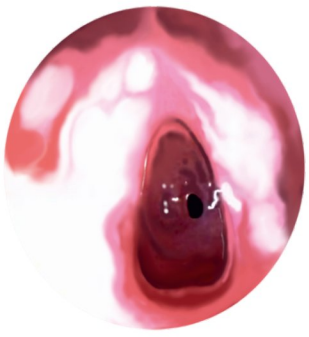
Treatment

- Unilateral: Wait and watch
  - One cord is paralyzed, but the other cord is still working
- Bilateral:
  - Tracheostomy (to get access to the airway)
  - Vocal cord lateralization (creates airway space for respiration)

SUBGLOTTIC STENOSIS

- **Definition:** Excessive thickening of the cricoid cartilage leading to narrowing of the subglottic lumen
  - The subglottis lies at the level of the cricoid cartilage
  - The subglottis is the narrowest portion of the larynx in children
- Subglottis diameter is abnormal if:
  - < 4 mm in full-term neonates
  - < 3 mm in premature neonates

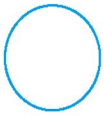
NEET PG 2021



Cotton-Myers Classification

- Used for classifying acquired subglottic stenosis, but applicable to congenital stenosis too.
  - **Grade 1**: obstruction up to 50%
  - **Grade 2**: 51-70% obstruction
  - **Grade 3**: 71-99% obstruction

- **Grade 4**: complete obstruction (100%)

Classification	From	To	Endoscopic appearance
Grade I	No Obstruction 	50% Obstruction	
Grade Il	51%	70%	
Grade Ill	71%	99%	
Grade IV	No Detectable Lumen		

Clinical Presentation

- Do not present at birth.
- The airway is narrowed but not completely blocked, resulting in minimal or very minor symptoms, which may include noisy breathing.
- Crying is essentially normal.
- Effect of crying: No increase in stridor

EFFECT OF CRYING

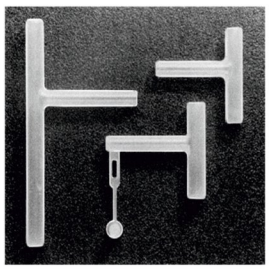
Condition	Crying effect on Stridor
Laryngomalacia	Increase/Worsen
Acute epiglottitis	
Congenital vocal cord paralysis	

Subglotic stenosis	No effect
Choanal atresia bilateral	Improve

Treatment Based On Grade

Grade	Management	Key Concept / Procedure
Grade 1	Observation	Usually resolves with growth; cartilages enlarge/expand with age
Grade 2	Endoscopic dilatation	Dilatation of the narrowed airway using endoscopic techniques
Grade 3	Laryngotracheal reconstruction	Rib cartilage graft used to expand the subglottic region
Grade 4	Partial Cricotracheal resection	Remove cricoid cartilage and 1st tracheal ring; connect larynx with trachea.

- Post surgery
  - Montgomery t-tube (Stent): Used in Grade 3 and Grade 4 reconstructions
  - Prevents further stenosis after surgery
    - Mitomycin-c (Antiproliferative drug)
  - Prevents fibrosis and adhesions

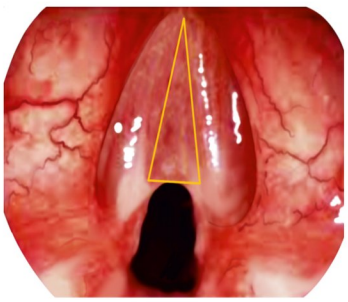


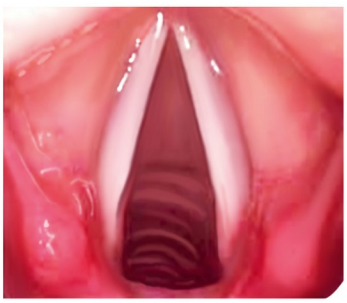
LARYNGEAL WEB

- Embryological Cause: Incomplete canalization of the larynx at the level of the vocal cords
- M/c site: At the level of the glottis

Types

- Partial web:
  - Only partially occupies the anterior 2/3 of the vocal cord
    → Weak cry
    → Noisy/Difficulty breathing
- Complete web:
  - It occupies the vocal cord completely.
  - Presents with no cry
  - The effect of crying will increase the stridors.





Treatment

- Partial web: Not an emergency
  - Eventually release the web
- Complete web: Emergency
  - Tracheostomy (to get access to the airway)
  - After tracheostomy → release the web

Common Error Files

01:05:08

- Conditions presenting at birth:
  - Laryngomalacia
  - Congenital vocal cord paralysis
  - Complete laryngeal web
- Stridor types:
  - Inspiratory: Laryngomalacia
  - Biphasic: Congenital vocal cord paralysis, laryngeal web
- Cry status:
  - Normal: Laryngomalacia, subglottic stenosis
  - Weak or absent: Congenital vocal cord paralysis, laryngeal web
- Effect of crying on stridor:
  - Increases: Laryngomalacia, Vocal cord paralysis, Laryngeal web, Epiglottitis
  - No effect: Subglottic stenosis
  - Improves: Croup/laryngotracheitis
- Effect of position: Seen in Laryngomalacia
  - Supine → worsens
  - Prone → improves
- Classification system:
  - Cotton-Meyer's classification is used for subglottic stenosis

INFLAMMATORY CONDITIONS OF LARYNX

01:09:54

Acute Epiglottis

- M/c in the age group: 3-7 years
- Causative organisms
  - Streptococcus pneumoniae - M/c
  - Haemophilus influenzae - Incidence is reduced due to vaccination
- Clinical presentation
  - Dysphagia
  - Odynophagia with drooling of saliva → dehydration
  - Absence of cough / Hoarseness of voice
  - Dyspnoea and Inspiratory stridor
    → Due to obstruction in supraglottic area
  - Increased chance of Laryngospasm on examination
  - Child sits in tripod position

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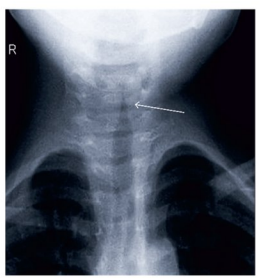
- Investigations
  - X-ray lateral view → Thumb sign → Inflammation of epiglottis
- Treatment
  - Secure the airways
  - Antibiotics against specific organisms
  - Steroids can be given to reduce the edema and to prevent airway obstruction
  - Voice rest and cough suppressants are not required
  - Racemic adrenaline nebulization or injection → Bronchodilator
  - Hydration is important
  - Intubation or tracheostomy may be required

Croup

- A/K/A Laryngo - Tracheo-Bronchitis
- M/c in the age group 3 months - 3 years
- Causative organisms
  - Parainfluenza virus type I and II (M/c) → Secondary infection by bacteria
  - Respiratory syncytial virus
  - Rhinovirus
  - Influenza virus
- Clinical presentation
  - H/o cold /URTI
  - Barking seal type of cough
  - Hoarseness of voice / change in cry is present
  - Sore throat absent
  - Dyspnoea and Biphasic stridor
    → Due to obstruction of sub-glottic area
  - Increased chance of Laryngospasm on examination
- Investigations
  - X-ray AP view → Steeple sign → narrowing of subglottis
- Treatment
  - Secure the airways
  - Antibiotics for secondary infections
  - Steroids can be given to reduce the edema
  - Racemic adrenaline nebulization with Helium and oxygen → Bronchodilator
  - Voice rest and cough suppressants are required
  - Hydration is important
  - Intubation or tracheostomy may be required



Q. In acute epiglottis in children, death occurs due to?
a. Atelectasis
b. Aspiration

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c. Laryngospasm
d. Airway obstruction

Ans: d

- Cause of death during examination → Laryngospasm

Q. A A 4-year-old child presented with sore throat, difficulty in swallowing, and difficulty in breathing. The child has a high-grade fever with tachypnoea. There is an inspiratory stridor with a recession of suprasternal and intercostal spaces. No strider is observed during expiration. The clinical picture is suggestive of:

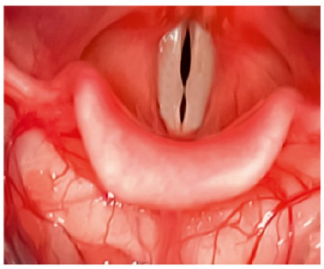
a. Bronchopneumonia
b. Lower tracheal obstruction
c. Intrathoracic obstruction
d. Extrathoracic obstruction

Ans: d

BENIGN LESIONS OF LARYNX

01:18:33

Vocal Nodule



- A/K/A Teacher's nodule / Singer's nodule
- B/L symmetrical lesions on the vocal cords
- Location: At the junction of anterior 1/3rd and middle 1/3rd
- Occurs due to gradual or chronic voice abuse
- Treatment
  - Voice rest
  - Microlaryngeal surgery

Vocal Polyp

- U/L lesion on the vocal cords
- Locations: At the junction of anterior 1/3rd and middle 1/3rd
- Occurs due to sudden voice abuse
- Presents with Diplophonia
- Treatment
  - Microlaryngeal surgery

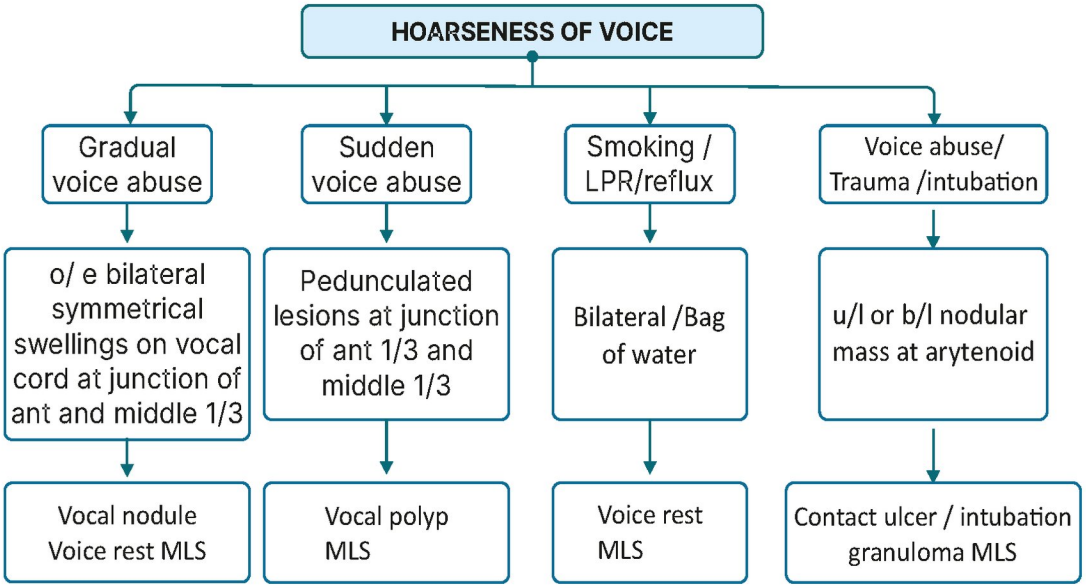
Reinke's Edema

- B/L symmetrical swelling of the vocal cords
- Location: At membranous or anterior 2/3rd of the vocal cords
- Cause - Smoking and GERD
- Presents with Diplophonia and low-pitch voice
- Appears as a bag of water
- Treatment
  - Voice rest
  - Antireflux therapy
  - Decortication / Stripping of the vocal cord epithelium
    → Initially, one vocal cord after 4 - 6weeks another vocal cord

→ Stripping of the vocal cord is not done simultaneously because it leads to adhesions of the vocal cord

Important Information

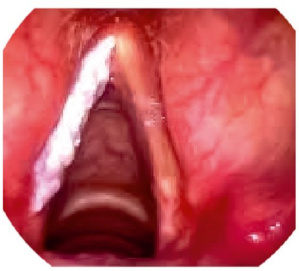
- Hoarseness is rough/harsh quality of voice
- Dysphonia is difficulty in speaking or any impairment of voice



Contact Ulcer

- A/K/A Pachyderma Laryngitis
- Cause - Trauma, reflux, and voice abuse → Hypertrophy of epithelium on one side with depression of another epithelium
- Pseudoulcer
- Affects the posterior part of the vocal cord

Keratosis Larynx



- A/K/A Leucoplakia

Smoking, Tobacco chewing
↓
Chronic irritation
↓
Transformation of epithelium to white plaque-like lesions

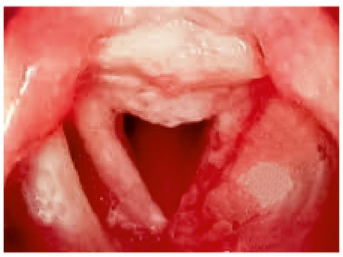
- Treatment
  - Stop the offending agents
  - Decortication/stripping of vocal cord or excisional biopsy of the vocal cord

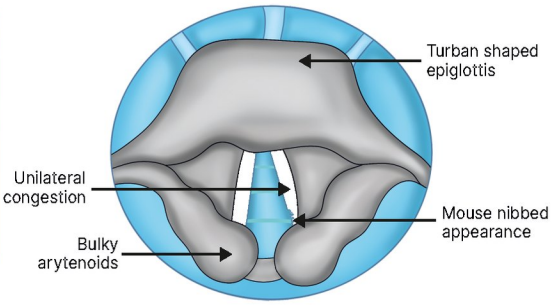
TB LARYNX

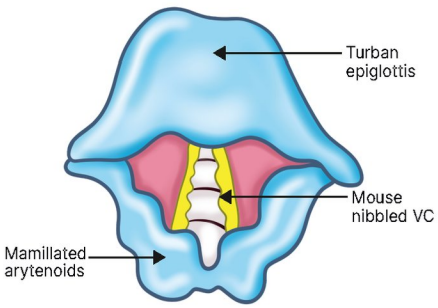
01:24:37

- It can be a/w with pulmonary tuberculosis
- It can also occur as low grade indolent form → Lupus Larynx
- Symptoms
  - Presents as painful ulcer → Odynophagia
  - Low-grade fever, cough, and weight loss

- Signs
  - Starts in the posterior part of the vocal cord
  - Hyperemia / Redness of the vocal cord with sluggish movement of vocal cord
  - Discontinuous ulcer → Moth-eaten / Mouse-nibbled
  - Swelling of arytenoids → Mamillated arytenoids
  - Swelling of epiglottis → Turban epiglottis







- Investigations
  - HPE → Caseating granuloma
  - CBNAAT
  - GeneXpert
  - Culture on LJ medium
- Treatment
  - Antitubercular drugs

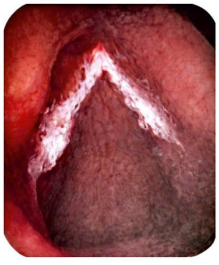
	TB Larynx	Lupus Larynx
Pulmonary TB	Yes	No
Activity of TB	Active	No
Parts affected	Posterior	Anterior
Symptomatic	Yes	No

Conditions of Larynx Presenting with Hoarseness

Clinical History
Hoarseness in an elderly with H/o smoking results Laryngopharyngeal reflux

Next step

Examination finding
O/E-white patches on both vocal cords.



Clinical History
Elderly, chronic smoker, presented with the complain of Hoarseness for one month and above

Next step

Examination finding
O/E-Ulceroproliferative growth on the TVC.

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Diagnosis
Laryngeal keratosis/ Leukoplakia
Premalignant

Management
Cessation of smoking excision by
MLS and biopsy.

Diagnosis
Biopsy-Squamous, cell Carcinoma

Investigations
- Neck examination for LN and crepitus
- CT to see tumor extent, spread and
- MRI best for seeing cartilage invasion

One Liners

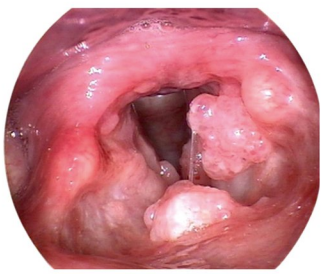
- Vocal nodules and polyps occur at junction of Anterior 1/3rd and Middle 1/3rd
- Sessile is Nodule and pedunculated is Polyp
- Diplophonia is seen in Vocal polyp / Reinke's edema
- Microlaryngeal surgery is the treatment of choice in Vocal polyp
- Voice rest is the treatment of choice in Vocal nodule
- Bilateral and symmetrical lesions are Vocal nodule
- Ulcer on one side and heaped up epithelium on the other is seen in Contact ulcer
- Microlaryngeal surgery involves using a microscope to excise the lesion with a focal length of 400nm
- Pachyderma laryngitis involves posterior part of vocal cords
- Not a true ulcer is Contact ulcer
- Painful laryngeal condition is TB
- Premalignant condition is Keratosis
- Stripping of vocal cord is done in Reinke's edema

Error Files

- Diplophonia → Vocal polyp
- Low-pitched voice → Reinke's edema
- Bilateral lesions
  - Vocal nodule
  - Reinke's edema
- Unilateral lesions:
  - Vocal polyp
  - Vocal cyst
  - Contact ulcer
  - Leukoplakia
- Condition involving the posterior part of the larynx:
  - Pachyderma laryngitis / Contact ulcer

LARYNGEAL PAPILLOMA

- Causative organisms
  - HPV 6 and 11
- 2 Types
  - Juvenile onset
  - Adult onset



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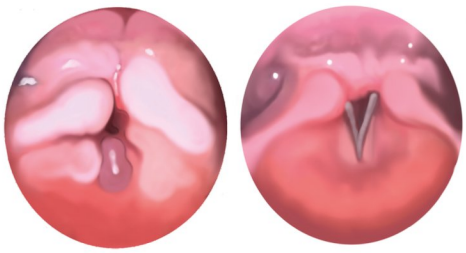
NEET PG 2019
FMGE 2023

	Juvenile papilloma's	Adult onset papilloma's
Age	Infants and young children	30-50years
Sex		It is twice more common in male
Number	Multiple lesion	Single lesion
Location	Multiple subsites -True and false vocal cords and epiglottis	Single subsite - Anterior half of vocal cord or anterior commissure
Aggressive	Aggressive, recurrence is common after surgical excision	Less aggressive, does not recur after excision
Recurrence	High	Less

- Treatment
  - Surgical excision using Microdebrider, coblation
  - Surgical excision with laser is not done
    → Because it generates fumes which mix with papilloma virus and forms Plumes which move to the lower trachea and cause distal seedling →Tracheal papilloma and Bronchial papilloma

VOICE AND SPEECH DISORDERS

Dysphonia Plica Ventricularis



- Voice is being generated by false vocal cords
- Cause
  - True vocal cord pathology - Inability of the true vocal cords to adduct
  - Psychological condition
  - Voice abuse
- Hypertrophy of the false vocal cord
- Treatment
  - Speech therapy
  - Reduction of the false vocal cord

Spasmodic Dysphonia

- Spasm of the phonatory muscle
- Cause
  - Neurological Disorder
    → Parkinsonism disease
    → Alzheimer's disease
  - Associated with other dystonias like Oromandibular dystonia and Blepharospasm

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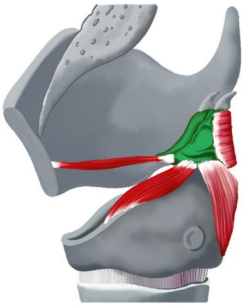
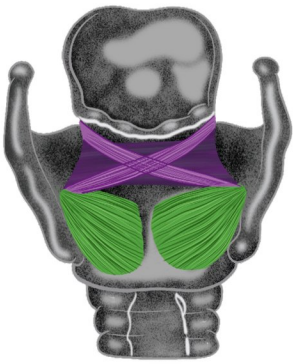
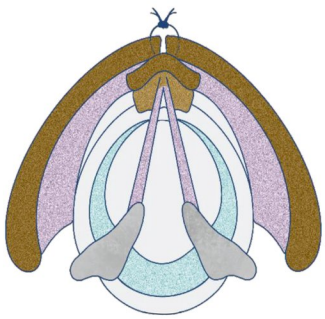
Types of Dysphonia		
Adductor dysphonia	**Abductor dysphonia**	**Mixed dysphonia**
• Spasm of vocalis muscle (medial fibres of the thyroarytenoid) • Scratchy, Croaky, strangulated type of voice 	• Spasm of posterior cricoarytenoid • Breathy type of voice is produced 	• Adductor + Abductor dystonia

- Treatment
  - Adductor dysphonia - Inj.Botox in thyroarytenoids muscle
  - Abductor dysphonia - Inj.Botox in cricoarytenoid muscle

Puberphonia



- Normally, in males during puberty, a transformation of voice occurs to produce an adult quality of voice, which is low-pitched
- In puberphonia, there is high pitch voice
- Pathology:
  - It occurs due to no correlation between development of the thyroid cartilage and vocal cord

Thyroid cartilage develops at a faster rate in comparison to vocal cord

↓

As a result vocal cord is pulled and stretched

↓

Produce a female quality of voice

- Diagnosis
  - Gutzmann pressure test

Applying pressure on the thyroid cartilage

↓

The vocal cord is relaxed

↓

Improvement in voice is seen → Male quality of voice produced

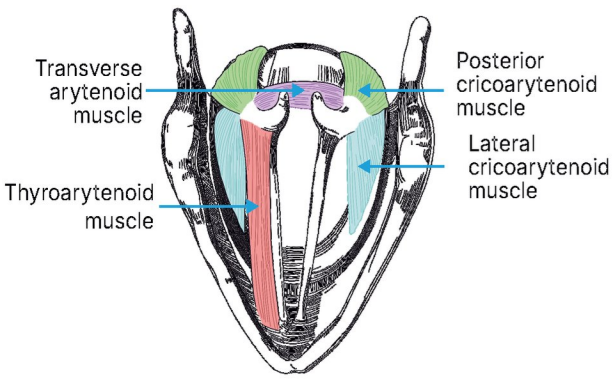
- Treatment
  - Speech therapy
  - Type III thyroplasty → Shortening the vocal cord → Release the tension on vocal cord

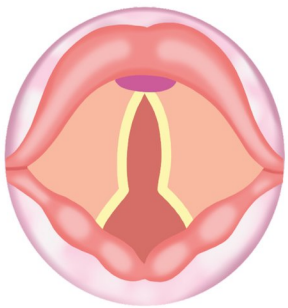
Important Information

- Type I Thyroplasty - Medialization of the vocal cord
- Type II Thyroplasty - Lateralization of the vocal cord
- Type III Thyroplasty - Shortening of the vocal cord
- Type IV Thyroplasty - Lengthening of the vocal cord

Phonasthenia

- Phonasthenia means weakness of the phonatory muscle
- The vocal cord lies between thyroid cartilage and the arytenoid cartilage
- Muscles between thyroid and the arytenoid cartilage → Thyroarytenoid
- Muscles between the 2 arytenoid cartilages → Interarytenoid muscle
- Keyhole appearance of the glottis
  - Weakness of Thyroarytenoid muscle → Anterior bowing of vocal cord
  - Weakness of Interarytenoid muscle → Posterior bowing of vocal cord





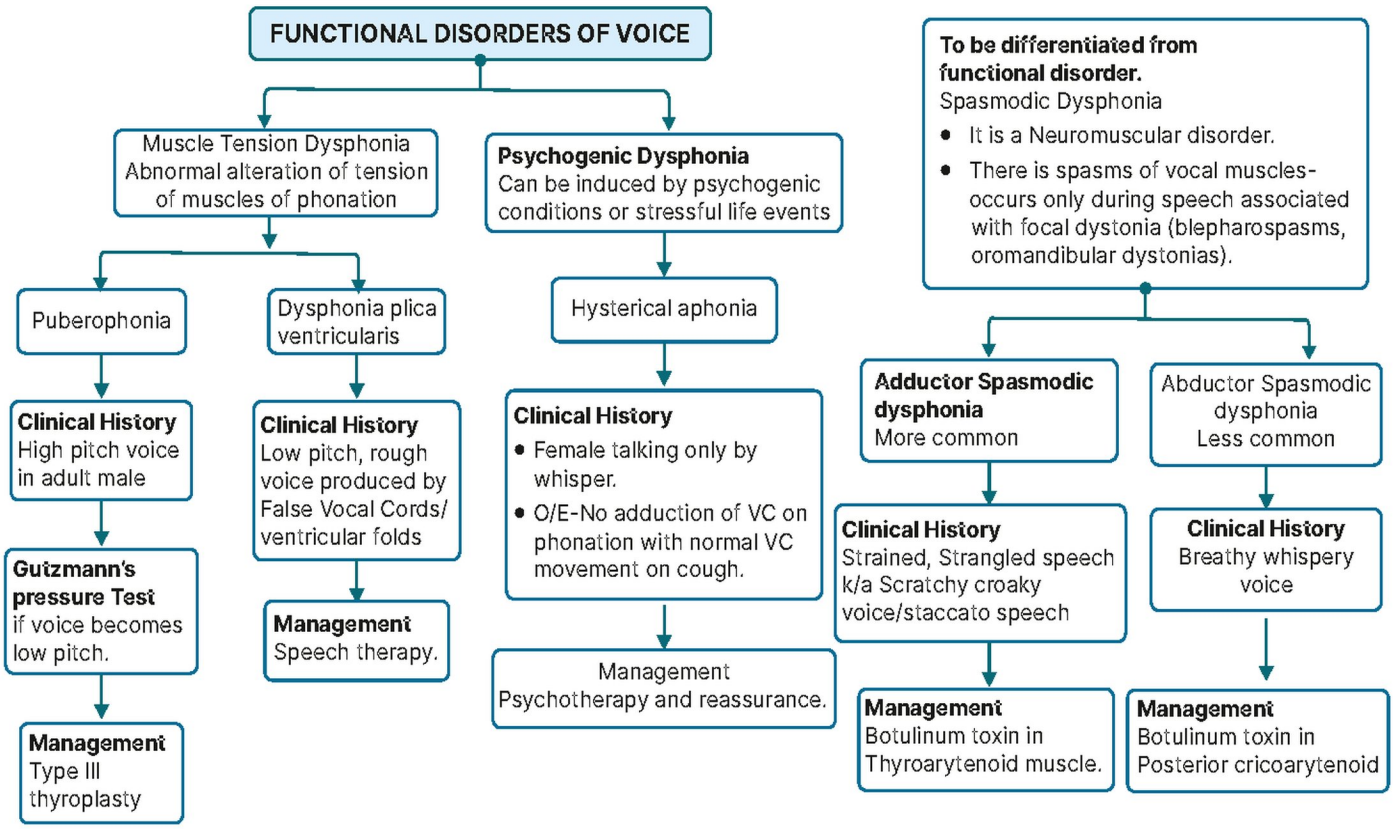
Functional Aphonia

- A/K/A Hysterical aphonia
- Seen in emotionally labile individuals
- Predisposing factors
  - Stress / Anxiety
  - Sudden emotional shock
- It is not associated with a neurological deficit
- Diagnosis

IDL examination / Laryngeal examination
↓
Asking the patient to cough
↓
Adduction of the vocal cord is seen
↓
Suggests vocal cord is mobile - Normal

- Treatment
  - Psychotherapy and Reassurance

YOURWISH



One Liners

- Gutzmann pressure test is used for the diagnosis of Puberphonia
- Keyhole appearance of the vocal cord is seen in Phonasthenia
- Adductor spasmodic dysphonia occurs due to Thyroarytenoid spasm
- Abductor spasmodic dysphonia occurs due to Posterior cricoarytenoid spasm

VOCAL CORD PARALYSIS

Superior Laryngeal Nerve

At the upper half of the neck
↓
At the level of hyoid bone - between bifurcation of carotid
↓
SLN divides into

External laryngeal nerve → Motor supply to circothyroid muscle

Internal laryngeal nerve
↓
Pierces thyrohyoid membrane in the upper part of the neck
↓
Sensory supply of larynx above vocal cord

Recurrent Laryngeal Nerve

- In right side → given at the level of the right subclavian artery

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- In left side → given at the level of the arch of aorta

Important Information

- Left recurrent laryngeal nerve is the only nerve enters the mediastinum

Nerve Injury In Relation To Thyroid Surgery

- In thyroid surgery, M/c affected RLN is Right
- Overall, M/c injured RLN is the left because of its longer course
- External laryngeal nerve can be injured while ligating the superior thyroid artery
- Recurrent laryngeal nerve can be injured while ligating the inferior thyroid artery
- In thyroid surgery, injury to the external laryngeal nerve > the recurrent laryngeal nerve

Nerve Supply Of Larynx

- Sensory
  - Supraglottis - Internal laryngeal nerve
  - Glottis - Recurrent laryngeal nerve
  - Subglottis - Recurrent laryngeal nerve
- Motor
  - All muscles except cricothyroid - Recurrent laryngeal nerve
  - Cricothyroid - External branch of the laryngeal nerve

Nerve Palsy

Nerve	Supply	Effect of injury / Palsy
ILN	Sensory to Supraglottis	• Aspiration
ELN	Supplies cricothyroid muscle, which is tensor and adductor of vocal cord	• Inability to raise the pitch
SLN	ILN+ELN	• Aspiration • Inability to raise the pitch
RLN	All muscles except cricothyroid	• Position of vocal cord - Median / Paramedian (Wagner and Grossman theory) - K/a - Abductor / Incomplete palsy • U/L palsy - Asymptomatic / Hoarseness • B/L palsy - Stridor
SLN+RLN	Complete palsy	• Complete / adductor palsy • Position of vocal cord - Cadaveric/intermediate position

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Clinical Significance Of Course Of Vagus

Surgeries in which areas	Can injure which branch to the larynx
Base of the skull	SLN,RLN,Complete palsy
Neck (carotid triangle)	SLN and its branches ELN , ILN
Thyroid triangle	Upper pole - ELN Lower pole - RLN
Mediastinal (Bronchogenic Ca , Aneurysm of arch of aorta , cardiovascular surgery - Bypass and grafting)	Left RLN

Management

- U/L RLN - Conservative management
- B/L RLN - Stridor - Tracheostomy followed by Lateralization by Cordectomy, Woodman's, Kashima Thyroplasty type II.
- Complete - Aphonia and aspiration - Medialization by Teflon, Fat, hydroxyapatite injection, thyroplasty type I

Position Of Vocal Cord In Health & Disease

Position	Location of cord from midline	Situations in	
		Health	Disease
Median	Midline / paramedian	Phonation	RLN paralysis
Intermediate (Cadaveric)	3.5mm This is the neutral position of the cricoarytenoid joint. Abduction takes place from this position	Maintained by the pull of Posterior cricoarytenoid, supplied by RLN, which pulls away from midline, and Cricothyroid, supplied by SLN, which pulls toward the midline	Combined paralysis of both RLN and SLN
Gentle abduction	7mm	Quiet respiration	Paralysis of adductors
Full abduction	9.5mm	Deep inspiration	

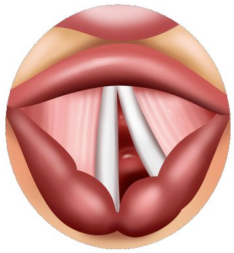
One Liners

- Sensory innervation below the vocal cord is RLN
- Sensory innervation above the vocal cord is SLN (ILN)
- Sensory anastomoses between ILN and RLN is called Galen's anastomoses
- All muscles receive motor innervation from RLN except Cricothyroid
- M/c cause of RLN palsy is Idiopathic >Traumatic
- M/c cause of RLN palsy on left side is Bronchogenic Ca, Aneurysm of aorta, mediastinal pathology

LARYNGEAL NERVE PALSIES

02:00:00

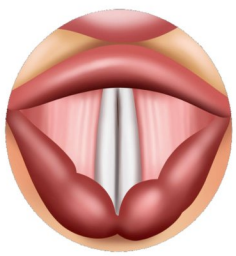
UNILATERAL RECURRENT LARYNGEAL NERVE (RLN) PALSY



Example: Right Recurrent Laryngeal Nerve Palsy

- Sensory Loss:
  - Right side of larynx, below the level of vocal cords
    → The recurrent laryngeal nerve is sensory below the level of the vocal cord
  - Left side: No sensory loss (because of unilateral palsy)
- Motor Loss
  - Right Side (Affected Side):
    → All muscles are paralyzed except the cricothyroid muscle
      → Cricothyroid is spared because it is supplied by the external branch of the superior laryngeal nerve (SLN)
  - Left Side (Normal Side):
    → All muscles functional (no palsy)
- Position of Vocal Cord
  - Normal position: Intermediate position (cadaveric position)
  - In RLN palsy:
    → Posterior cricoarytenoid is paralyzed → abduction function lost
    → Adduction and tension function is retained
    → The paralyzed cord lies in the median or paramedian position
- Effects on Phonation → No major change
  - Paralyzed cord already in median/paramedian position
  - Opposite cord can come to the midline
  - Both cricothyroids are functional
  - Adduction happening normally
  - May have mild change in voice due to paralysis of other adductors
- Effects on Respiration
  - Paralyzed cord cannot move away from midline (abductor paralyzed)
  - Normal cord can move away from midline
  - Result: No stridor, but dyspnea on exertion present
- Treatment: Wait and watch

BILATERAL RECURRENT LARYNGEAL NERVE (RLN) PALSY



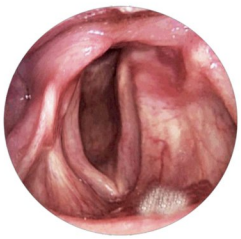
- Sensory Loss
  - B/L sensory loss below the level of the vocal cords
- Motor Loss
  - Both posterior cricoarytenoids affected and paralyzed
  - Bilateral paralysis of all muscles except cricothyroid
- Position of Vocal Cord
  - Both vocal cords lie in the midline or paramedian position
    → Abductor is paralyzed on both sides
- Effects on Phonation
  - Both cords are already in the midline
  - Cricothyroid is functional

YOURWISH

- Result: No major change in voice
- Some change may occur due to paralysis of other adductors (thyroarytenoid, interarytenoid)

- Effects on Respiration
  - Both vocal cords are unable to move away from the midline → Stridor
    → As B/L Posterior cricoarytenoid is paralyzed
  - Stridor is a feature of bilateral RLN palsy
- Treatment
  - Immediate: Tracheostomy
  - After tracheostomy: Create space for air
    → Lateralize the cord
    → Resect some tissue to create space
  - Surgical Procedures (Type 2 Thyroplasty):
    → Arytenoidectomy: Remove only the arytenoid to create space
    → Cordectomy: Remove only the posterior 2/3rd of the vocal cord
    → Kashima's Operation: Remove arytenoid + posterior 2/3rd of cord

UNILATERAL SUPERIOR LARYNGEAL NERVE (SLN) PALSY



- Superior laryngeal nerve → Internal laryngeal nerve
  - Sensory to the larynx above the level of vocal cord

Example: Right Superior Laryngeal nerve palsy

- Sensory Loss:
  - Right side: Sensory loss above the level of vocal cords
  - Left side: Normal sensation
- Motor Loss:
  - Right Side (Affected):
    → Only the cricothyroid muscle is paralyzed
    → Rest of muscles are functional (supplied by recurrent laryngeal nerve)
  - Left Side:
    → All muscles functional (unilateral SLN palsy)
- Position of Vocal Cord:
  - The adductor and tensor (cricothyroid) are paralyzed
    → Posterior cricoarytenoid has more power
    → Pulls the vocal cord to the abducted position
  - ASKEW Position:
    → Normal cord in intermediate position + Paralyzed cord in abducted position
    → Seen in: Unilateral SLN palsy because the cricothyroid is paralyzed on that side.
- Effects on Phonation
  - Paralyzed cord cannot go to the midline, as the cricothyroid is paralyzed
  - Other adductors can pull it, but cannot do the tensor function
  - Result: Definite change in voice → Cannot increase pitch or tone
- Effects on Respiration
  - Paralyzed cord already in the abducted position
  - Normal cord can go to the abducted position
  - Result: No effect on respiration

- Moderate risk of aspiration
- Treatment
  - Aim: Reduce aspiration
  - Procedure: Medialize the cord → Type 1 Thyroplasty
    → Put fat graft in the paraglottic space
    → Push the cord medially
    → Fat injection in the vocal cord

BILATERAL SUPERIOR LARYNGEAL NERVE (SLN) PALSY

- Sensory Loss:
  - Bilateral sensory loss above the level of the vocal cord
- Motor Loss
  - Bilateral cricothyroid muscles are paralyzed
  - Rest of muscles are functional
- Position of Vocal Cord
  - Posterior cricoarytenoid has more power → Vocal cord in abducted position
- Effects on Phonation
  - Cannot come to midline, cannot stretch
  - Aphonia is a feature of bilateral SLN palsy
- Effects on Respiration
  - Both paralyzed cords in the abducted position
  - Result: No problem with respiration
- Risk of Aspiration
  - Highest risk of aspiration
    → Bilateral sensory loss above the level of the vocal cords
    → Both cords are 7mm away from midline → Total 14mm gap
- Treatment
  - Initially: Tracheostomy (protect airway)
  - For intractable aspiration: Tracheoesophageal diversion + Permanent tracheostomy
    → Divert 1st tracheal ring with the upper esophagus
    → Any aspiration through larynx → goes through the 1st tracheal ring into the esophagus
    → From the 2nd tracheal ring → bring out as a permanent tracheostomy

UNILATERAL COMBINED PALSY (SLN + RLN on ONE side)

Example: Right side SLN + RLN palsy

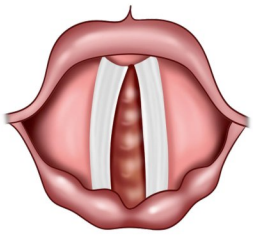
- Sensory Loss
  - Right hemilarynx: Anesthetized
  - Left hemilarynx: Completely functional
- Motor Loss
  - Right Side: All muscles are paralyzed
  - Left Side: All muscles functional
- Position of Vocal Cord
  - Affected cord → Intermediate position with bowing
  - Opposite cord → Normal intermediate position
- Effects on Phonation

- Difficulty in speaking
- Mild change in voice

- Effects on Respiration
  - No stridor
  - Dyspnea on exertion +
- Moderate risk of aspiration
- Treatment:
  - Tracheostomy to protect the airway
  - Type 1 Thyroplasty: Medialize the paralyzed cord after tracheostomy

BILATERAL COMBINED PALSY (Bilateral SLN + RLN)

02:15:28



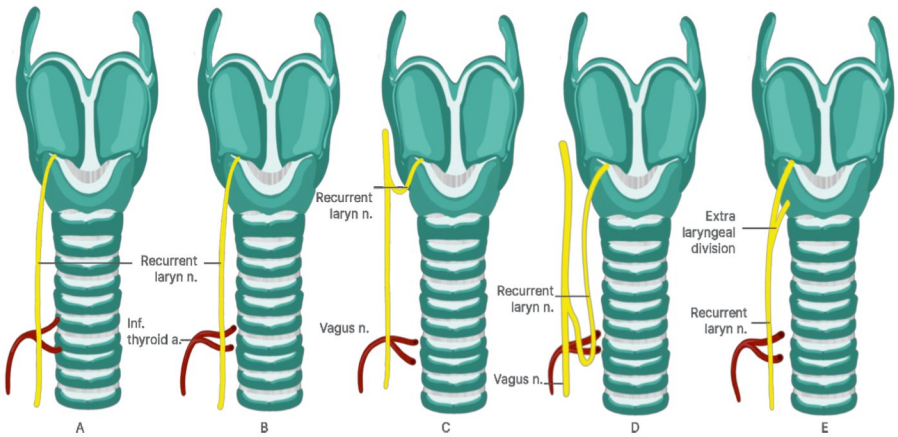
- Sensory Loss
  - Sensory loss in the complete larynx
- Motor Loss
  - All muscles are paralyzed bilaterally
- Position of Vocal Cord
  - Both cords are in intermediate position, but bowing
- Effects on Phonation
  - Both cords cannot come to the midline → aphonia
  - Aphonia is feature of bilateral SLN palsy and bilateral combined palsy
- Effects on Respiration
  - Dyspnea on exertion
  - Not stridor
- Risk of Aspiration
  - Complete sensory loss of larynx High risk of aspiration
- Treatment
  - Directed towards aspiration
  - Tracheostomy to protect airway

ANATOMICAL RELATIONS AND TRIANGLES

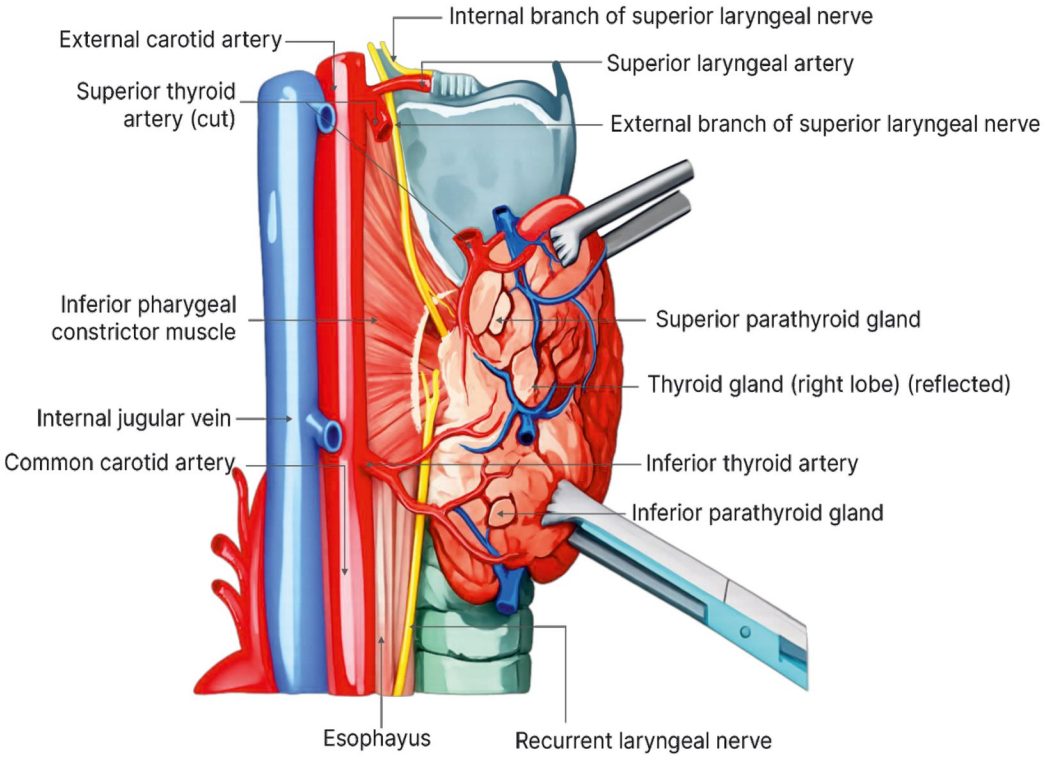
02:16:40

RELATION TO INFERIOR THYROID ARTERY

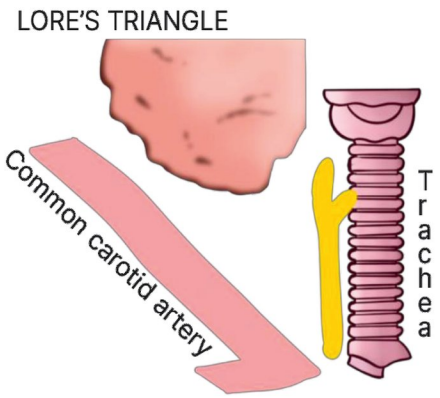
- Recurrent laryngeal nerve in close proximity to inferior thyroid artery
  - When ligating inferior thyroid artery on right side → High risk of recurrent laryngeal nerve palsy

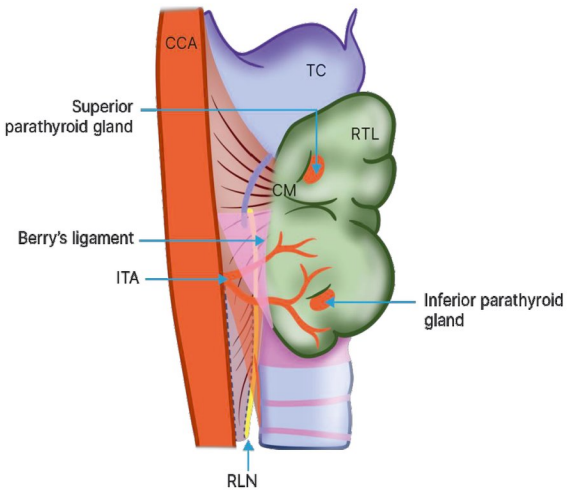


TRIANGLES ASSOCIATED WITH RECURRENT LARYNGEAL NERVE



- Common Landmarks :
  - Laterally: Common carotid artery
  - Medially: Recurrent laryngeal nerve in the tracheoesophageal groove
- BEAR'S TRIANGLE
  - Superior boundary: Inferior thyroid artery
- LOWRY'S TRIANGLE
  - Superior boundary: Lower pole of thyroid
- SIMON'S TRIANGLE
  - Superior boundary: Cricothyroid muscle





TRIANGLE ASSOCIATED WITH SUPERIOR LARYNGEAL NERVE

02:19:25

JOLLS'S TRIANGLE

- Boundaries:
  - Medially: Midline
  - Laterally: Upper pole of thyroid (with superior thyroid vessels and superior laryngeal vessels)

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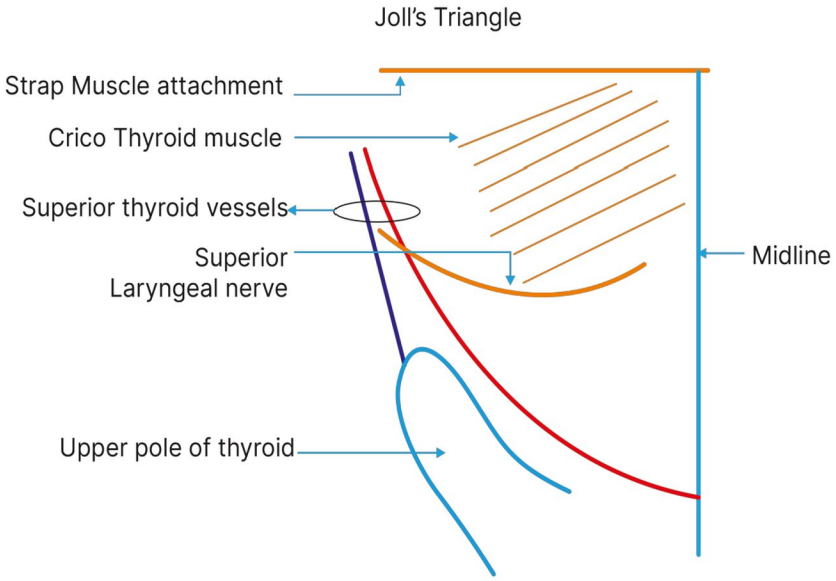
- Superiorly: Insertion of strap muscles
- Floor: Cricothyroid muscle (supplied by external branch of SLN)



Important Information

- Thyroid:
  - Lower pole related to the recurrent laryngeal nerve
  - Upper pole related to the external branch of SLN

ANESTHESIA FOR INTERNAL LARYNGEAL NERVE

- Injection given below the greater cornua of the hyoid bone
- The internal laryngeal nerve is found at this location
- Given off from the superior laryngeal nerve at this part → Pierces the thyrohyoid membrane
  - Gives sensation to the larynx above the level of vocal cord

ERROR FILES

- Strider:
  - Seen only in bilateral RLN palsy
- Aphonia is seen in:
  - Bilateral SLN palsy
  - Bilateral combined palsy
- Vocal cord position in midline: In RLN palsy
- Vocal cords in abducted position: In SLN palsy
- Askew position:
  - Paralyzed cord in abducted position and bowing
  - Normal cord in intermediate position
  - Seen in unilateral SLN palsy
- Highest risk of aspiration: Bilateral SLN palsy
  - Vocal cords are away from each other in abducted position
  - Bilateral loss of sensation above the level of vocal cord

YOURWISH

YOURWISH

LARYNGEAL CANCER

02:22:10

Feature	Supraglottis	Glottis	Subglottis
Incidence	Intermediate	Most common	Least common
Site	Most common in epiglottis	Free edge & superior surface of anterior & middle 1/3 of vocal cord	Midline in anterior half
Appearance	Suprahyoid - Exophytic Subhyoid - Ulcerative	Nodule/ulcer/thickening of the vocal cord	Submucosal nodule
Vocal cord involvement	Very late → hoarseness is a late feature	Early involvement → early hoarseness Fixation very late (bad prognostic sign) → Indicates infiltration of thyroarytenoid muscle	Late
Metastases	• Very early (rich lymphatics) • Bilateral nodes in epiglottic carcinoma (midline) • Upper & middle deep cervical nodes	None (scant lymphatics in vocal cords)	• Early • Bilateral • Lower deep cervical, pre-tracheal, para-tracheal
Symptoms	Silent until late Late: Dysphagia, throat pain, referred ear pain, respiratory distress	Hoarseness of voice	Silent Respiratory distress (late)
Nature	Anaplastic, highly malignant	Well differentiated, less malignant	—
Prognosis	Bad → Late symptoms + early lymphatic spread + high malignancy	Good → Early detection (HOV) + no lymphatic spread + well differentiated	

PYQ POINTS

- MCn carcinoma: Squamous cell carcinoma
- MC site is at the level of the glottis
- Gender: Usually seen in males
- Presentation: Hoarseness of voice
- Smokers, > 50, males, hoarseness of voice for > 3 weeks → evaluate for malignancy
- Lymphatic metastasis:
  - Least/no metastasis: Glottic cancers

YOURWISH

- Maximum lymphatic spread: Supraglottic malignancies
- Bilateral metastasis: Epiglottic cancer

- Pain in the ear in carcinoma larynx is referred via the vagus nerve
  - Vagus gives auricular branch called Arnold's nerve
- Laryngeal ligaments and membranes act as barriers to prevent tumor spread

TNM STAGING

02:24:50

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Stage	Criteria
Stage 1	T1, N0, M0
Stage 2	T2, N0, M0
Stage 3	T3, N0, M0 or T1/T2/T3+ N1, M0 Rule: N1 → Stage 3
Stage 4a	T4A, N0, M0 or T1/T2/T3/T4a + N2, M0 Rule: N2 → Stage 4A
Stage 4b	T4B, N0, M0 or T1/T2/T3/T4a/T4b + N3, M0 Rule: N3 → Stage 4B
Stage 4c	Any T, Any N, M1 Rule: Any metastasis → Stage 4C

T Stage	Description
T1	Tumor in one subsite of the larynx (glottis/supraglottis/subglottis) The vocal cord is fully mobile
T2	Tumor in >1 subsite of the larynx The vocal cord is completely or partially mobile
T3	Tumor in >1 subsite of the larynx ANY of: Vocal cord fixed Involves pre-epiglottic / paraglottic space Involves inner perichondrium
T4a (Local spread)	Tumor breaks inner + outer perichondrium Extends outside larynx: Anterior → Strap muscles Superior → Base of tongue Posterior → Hypopharynx Inferior → Trachea

T4b (Advanced/distant spread)	Extends to: Retropharyngeal/prevertebral space via hypopharynx Advanced unresectable spread

N Stage	Description
N0	No neck nodes
N1	Node < 3 cm
N2	Nodes 3-6 cm
N2a	Single ipsilateral node
N2b	Multiple ipsilateral nodes
N2c	Contralateral nodes
N3	Node > 6 cm
N3a	No extranodal spread
N3b	Extranodal spread present

M Stage	Description
M0	No distant metastasis
M1	Distant metastasis present

INVESTIGATIONS

- IOC: Biopsy
- First line investigation → Radiological: Contrast-enhanced CT scan
  - MRI is indicated only when cartilage or spaces around the larynx are involved

TREATMENT

STAGE 1 AND STAGE 2 (PRIMARY PATIENT)

- First-line treatment: Radiotherapy
  - Voice preserved
  - Exception:
    → Carcinoma of the middle 1/3rd of the vocal cord → Microlaryngeal surgery
- Recurrence in Stage 1 and 2
  - Cannot give radiation again Conservative surgery
    → Partial laryngectomy (based on lesion)

STAGE 3 OR EARLY 4A (PRIMARY)

- Treatment: Concurrent chemo-RT
  - Cisplatin/ Carboplatin/ Cetuximab

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- Large T3 or T4a with one arytenoid spared → Near total laryngectomy
  - Remove the entire larynx except the arytenoid not involved in the disease
- Advanced disease - B/L arytenoids involved → Total laryngectomy
- Stage T3 or T4a with recurrence → Total laryngectomy
- Patient cannot take chemo-rt (e.g., prior chemoradiation) → Total laryngectomy
- Carcinoma of the epiglottis spreading to the pre-epiglottic space:
  - If a small lesion on the epiglottis is just going to the pre-epiglottic space with no nodal spread → Partial laryngectomy

VOICE REHABILITATION (After Total Laryngectomy)

02:36:49

- Esophageal speech: Can speak only a few words
- Tracheoesophageal speech: Serviceable speech, can talk sentences
- Electrolarynx: Artificial larynx, not commonly used

NECK DISSECTION IN CARCINOMA LARYNX

02:37:05

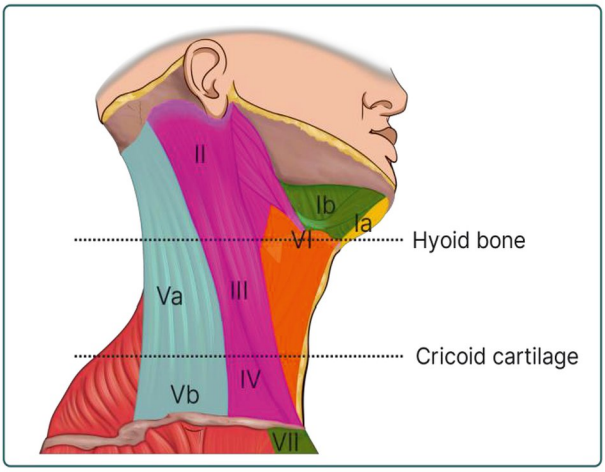
NECK DISSECTION IN CARCINOMA LARYNX

Indication of Neck Dissection in Carcinoma Larynx

- If any neck node is present, it should be managed by neck dissection if the primary is treated by surgery and by RT if the primary is treated by RT.
- In No neck-if chances of occult LN is > 15% →e.g. supraglottic Carcinoma → Selective lateral neck dissection (removal of only selective LN and no non lymphatic structures) should be done

Lateral neck dissection removes LN II, III and IV. Anterior compartment neck dissection removes level VI. (Delphian node)

Levels of Lymph Node in the Neck



Levels Ia : Submental
Level Ib : Submandibular
Level II : Upper deep cervical
Level III : Middle deep cervical
Level IV : Lower deep cervical

Levels Va : Posterior triangle
Level Vb : Supraclavicular
Level VI : Anterior compartment (lymph nodes)
Level VII : Superior mediastinal

NECK DISSECTION IN CARCINOMA LARYNX

Radical ND
LN removed-I to V
Nonlymphatic structures removed

- Nerve -Spinal accessory
- Vein- IJV
- Muscle- SCM
- tail of Parotid, submandibular salivary gland, omohyoid

Modified RND
LN removed-I to V
Structures preserved

- Spinal accessory nerve-MRND 1
- Spinal accessory nerve+IJV→ MRND II
- Spinal accessory nerve+IJV+SCM →MRND III also k/a funtional neck dissection/Comprehensive neck dissection

YOURWISH

YOURWISH

ERROR FILES – LARYNGEAL CANCER

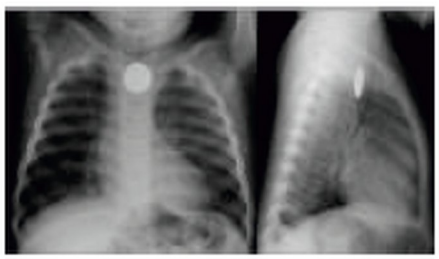
- Best prognosis: Glottic malignancies (early presentation + no lymphatic spread)
- T3 → 3 rules:
  - Vocal cord fixation
  - Spaces involved
  - Inner perichondrium involved
- First investigation: Contrast-enhanced CT scan
- Role of MRI: Only when cartilage or space involvement
- Total laryngectomy indications:
  - Bilateral arytenoid involvement in a primary patient with a large malignancy
  - Patient who cannot tolerate chemo-RT
  - Stage T3 or T4A with recurrence

FOREIGN BODY IN AIRWAY AND ESOPHAGUS

02:39:50

FOREIGN BODY IN TRACHEA

- X-ray AP view: Appears SLIT-LIKE rim
- X-ray lateral view: Appears ROUND



FOREIGN BODY IN UPPER ESOPHAGUS

- X-ray AP view: Appears ROUND
- X-ray lateral view: Appears SLIT-LIKE
- Airway seen anterior to the coin

FOREIGN BODY IN AIR PASSAGE

- Presentation:
  - Common in the younger age group (1-4 years)
  - MC FB → Nuts and peanuts
  - Earliest symptoms: Sudden onset, paroxysmal coughing, choking, gagging
  - Later: Hoarseness, hemoptysis, stridor

MANAGEMENT

FOREIGN BODY LARYNX

Management

First aid

If no resp distress, able to speak-Back blows and encourage coughing.

If Conscious with Universal choking sign (inability to speak, breathe, cough)→ Heimlich maneuver also k/a abdominal thrustes.

If F.B visible-finger sweep maneuver

Complications
Gastric and esophageal perforation, rib fracture, injury to abdominal organs.

Contraindication
- Age-< 1yr
- Unconscious-give sternal compression
- Pregnancy-give chest thrust (over the sternum).

IF FIRST AID FAILS:

- In CICO (Cannot intubate, cannot oxygenate)
  - Do emergency Cricothyrotomy - For temporary access to the airway
    ↓
  - Minitracheostomy – Vertical incision given in the cricothyroid membrane.
    ↓
  - Removal by rigid Laryngotracheobronchoscopy

Cricothyroid membrane
site of cricothyrotomy

HEIMLICH MANEUVER

- Patient held from behind, sudden inward and upward thrusts given over the epigastrium below the sternum
- To do when the patient is conscious & showing signs of distress

MCQ

Q1. A singer is finding difficulty in raising the pitch of her voice. On examination, her vocal cords were depressed and bowing. Which of the following muscles is involved?

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a. Cricothyroid
b. Interarytenoid
c. Posterior cricoarytenoid
d. Lateral arytenoid

Answer: **a**

Q2. Which of the statements is true regarding laryngomalacia?

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a. Expiratory stridor
b. Breathing improves in supine position
c. Most common congenital abnormality in infants
d. Surgical management is required in most children

Answer: **c**

Q3. A child is brought to the hospital with respiratory distress and biphasic stridor. The radiograph is shown below. What is the diagnosis?

a. Acute epiglottitis
b. Acute laryngotracheobronchitis
c. Foreign body aspiration
d. Laryngomalacia

Answer: **b**

Q4. A 55-year-old patient comes with hoarseness of voice and difficulty swallowing. The patient was diagnosed with laryngeal carcinoma, and surgical management was done. The post-operative image of the patient is given below. Which of the following surgery was done on this patient?

a. Partial laryngectomy
b. Percutaneous tracheostomy
c. Standard tracheostomy
d. Total laryngectomy

Answer: d

YOURWISH